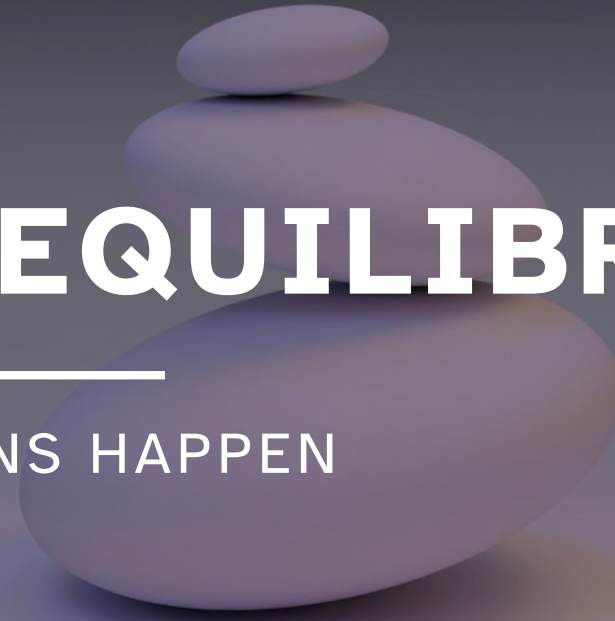


Three smooth, light-colored stones are stacked vertically in the center of the image. The bottom stone is the largest, the middle one is medium-sized, and the top one is the smallest. They are all perfectly balanced on top of each other.

15A CHEMICAL EQUILIBRIUM

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 15

1. Define and explain the condition of equilibrium.
2. Determine equilibrium constant expressions and their values.
3. Use equilibrium constant expressions as formulas.
4. Manipulate equilibrium constant expressions based on changes to the chemical equation.
5. Define and explain the reaction quotient.
6. Predict the direction of a reaction using the relationship between the reaction quotient and the equilibrium constant.
7. Use ICE tables to solve equilibrium problems.
8. Identify the reaction conditions that shift the equilibrium position of a reaction.

Introduction to Equilibrium

15.1

A **single arrow** ($\rightarrow$) in chemical equations implies that the reactants are *only* and *entirely* converted to products (i.e., a *complete forward* reaction).



Some reactions do go to completion, but many others do **NOT**.

A **double arrow** ($\rightleftharpoons$) is used to denote reactions that occur in *both directions* at the same time. These reactions *never* go to completion.

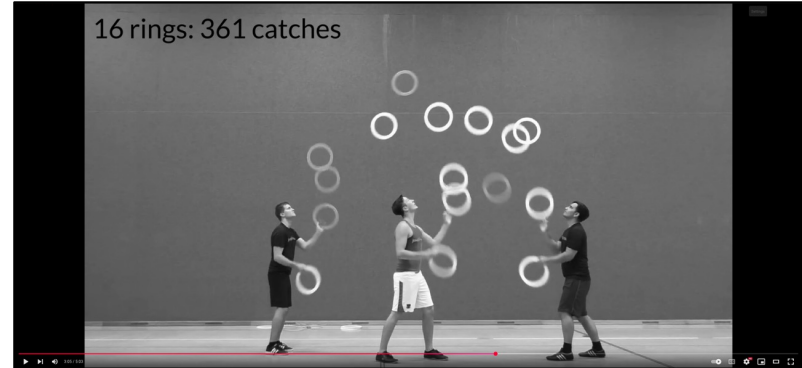


left-to-right is forward direction

right-to-left is reverse direction

Introduction to Equilibrium

15.1



Chemical equilibrium

The stage in a chemical reaction when there is **no further *net changes* in the composition** of the reaction mixture.

All chemical equilibria are dynamic equilibria!

- The forward reaction rate is equal to the reverse reaction rate.

$$\text{Rate}_{\text{for}} = \text{Rate}_{\text{rev}} \quad \text{kinetics!}$$

- Neither reaction stops at equilibrium.
- Both reactions continue with no *net change* in concentrations of products or reactants.

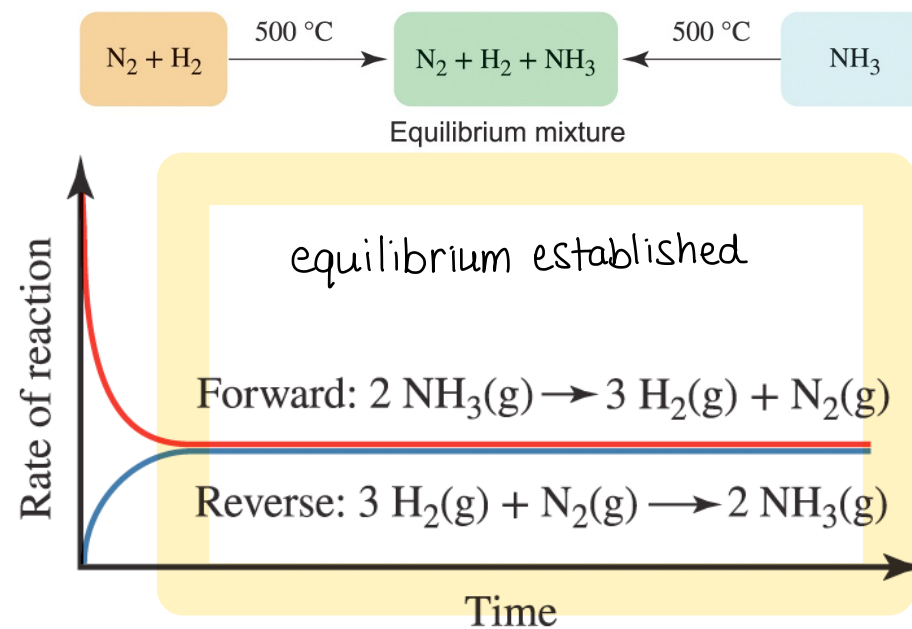


FIGURE 15.1 & FIGURE 15.2

Equilibrium Constants

15.2

Equilibrium constants (K) also called Law of Mass Action

For any reaction *at equilibrium*, K is the ratio of the concentrations of the products divided by the concentrations of the reactants, all of which are raised to their stoichiometric coefficients.



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

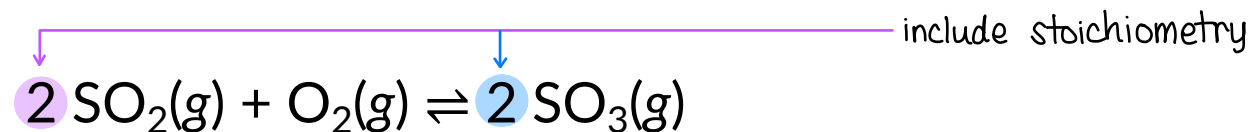
Important Notes about K

- Values of K depend on the specific **chemical equation** and on the **temperature**.
 - A reaction *at equilibrium* will always have the **same value** of K .
- Equilibrium constant expressions do **not** include any pure solids or pure liquids.
 - The concentrations of pure solids and liquids are constant.
 - Only **aqueous and gaseous species** appear in equilibrium constant expressions as only these species have concentrations that can change.
- All equilibrium constants (K) are **unitless** or **dimensionless**.

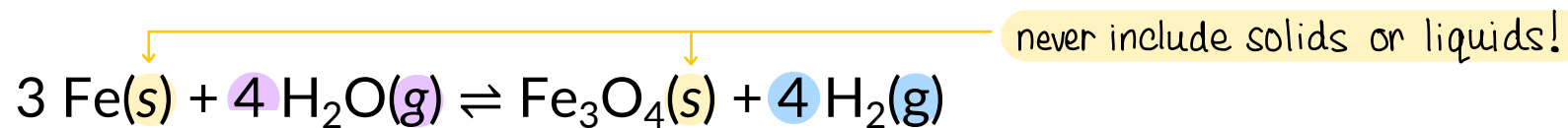
EXAMPLE PROBLEM

Learning Objective(s): 15.2

Write the equilibrium constant expressions (K) for the following two chemical reactions.



$$K = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]}$$



$$K = \frac{[\text{H}_2]^4}{[\text{H}_2\text{O}]^4}$$

Equilibrium Constants

Chemistry in Context

Many chemical reactions are equilibria with associated equilibrium constants (K).

Even though equilibrium constants are *unitless*, sometimes chemists choose to explicitly denote that *concentrations* are used by writing K_c instead of just K .



$$K \equiv K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

When *all* species are *gases*, K can *also* be expressed in terms of *partial pressures* to give K_p .



$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

For reactions that have the *same* number of moles of *gases* on *both sides* of the balanced chemical equation (e.g., $a + b = c + d$), these values are *equivalent*: $K \equiv K_c \equiv K_p$.

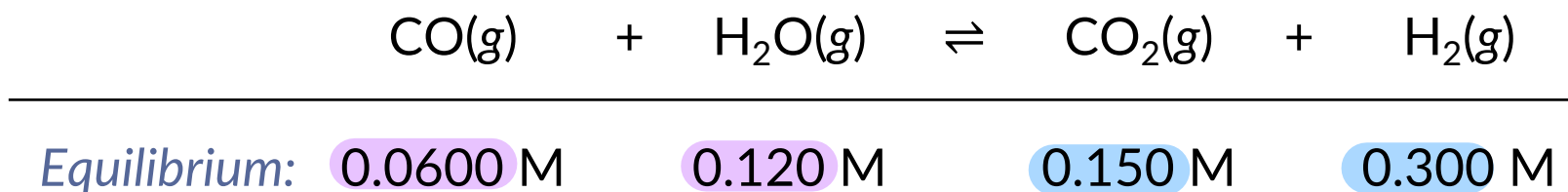
Not expected of you in this class: There is an equation to convert between K_c and K_p when $a + b \neq c + d$.

EXAMPLE PROBLEM

Learning Objective(s): 15.2

At a particular temperature, the following concentrations were obtained at equilibrium for the reaction given below.

Calculate the value of K_c for this reaction.



$$K = \frac{[\text{CO}_2][\text{H}_2]}{[\text{CO}][\text{H}_2\text{O}]} = \frac{(0.150)(0.300)}{(0.0600)(0.120)} = 6.25$$

note: no units!

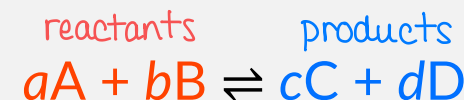
Equilibrium Constants

15.2

What does K tell us about the reaction?

The *magnitude* of K tells us whether the reaction (as it is written) is *product-favored* or *reactant-favored* at equilibrium.

We sometimes refer to this as the “*extent of the reaction*.”



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

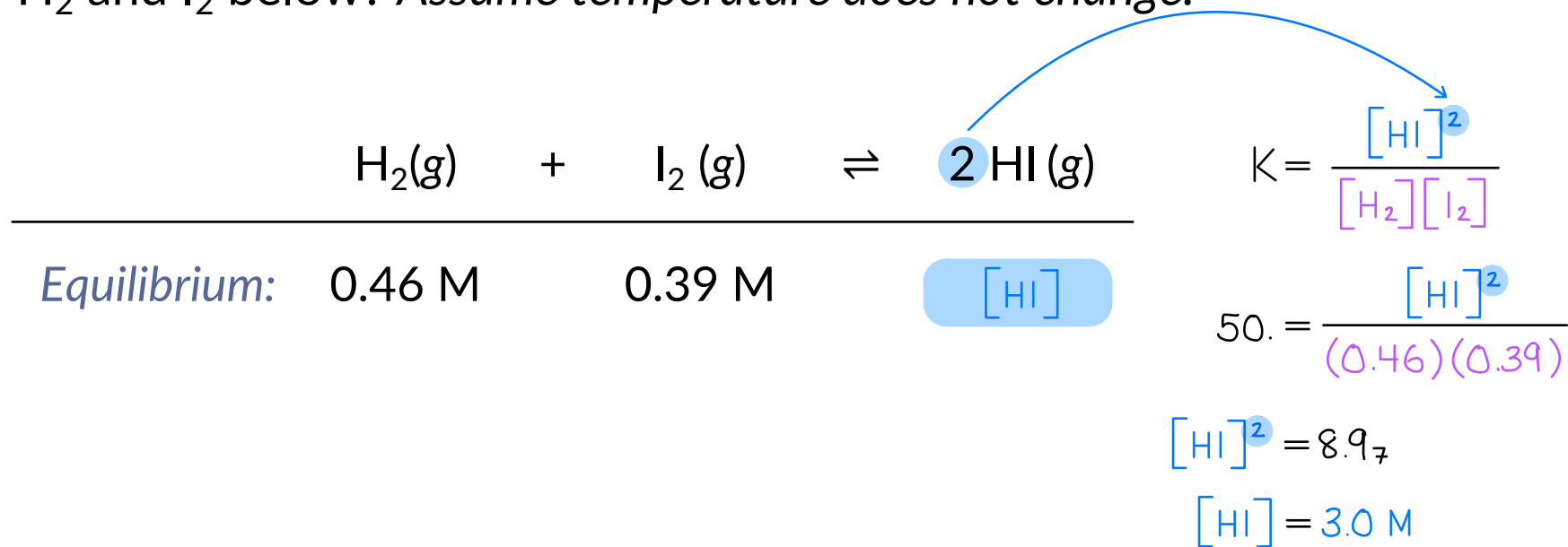
- Large values ($K \gg 1$): Equilibrium favors *products*; relatively *high* proportion of products-to-reactants; mostly *products*
- Intermediate values ($K \approx 1$): Equilibrium favors neither products nor reactants strongly; relatively *equal* proportion of products-to-reactants
- Small values ($K \ll 1$): Equilibrium favors *reactants*; relatively *low* proportion of products-to-reactants; mostly *reactants*

IN-CLASS QUESTION 1

Learning Objective(s): 15.2, 15.3

At a given temperature, the following reaction has a $K_c = 50$. note: no units!

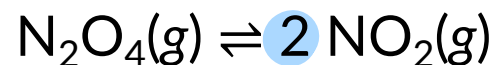
What is the equilibrium concentration (in molarity) of HI given the equilibrium concentrations for H_2 and I_2 below? *Assume temperature does not change.*



IN-CLASS QUESTION 2

Learning Objective(s): 15.2, 15.3

The equilibrium between N_2O_4 and NO_2 has a $K_p = 47.9$ at 400 K. note: no units!



A certain quantity of N_2O_4 is placed into a sealed container and the system is allowed to come to equilibrium at 400 K.

Calculate the partial pressure (in atm) of NO_2 in the sealed vessel if the partial pressure of N_2O_4 is found to be 0.335 atm.

$$K = \frac{(P_{\text{NO}_2})^2}{P_{\text{N}_2\text{O}_4}}$$

$$47.9 = \frac{(P_{\text{NO}_2})^2}{0.335}$$

$$(P_{\text{NO}_2})^2 = 16.05$$

$$P_{\text{NO}_2} = 4.01 \text{ atm}$$

Using Equilibrium Expressions

15.3

Manipulating equilibrium constants (K)

We typically read and write equilibrium constants (K) for the “*forward*” reaction. This is the reaction that occurs from *left-to-right* based on the balanced chemical equation written.



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

The “*reverse*” reaction is the reaction that occurs from *right-to-left* based on the *same* balanced chemical equation written previously.

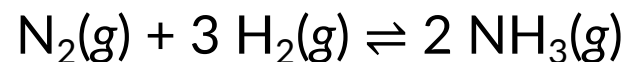
- The equilibrium constant for the *reverse* reaction (K') is always the *reciprocal* of the equilibrium constant for the corresponding forward reaction.



$$K' = \frac{[A]^a [B]^b}{[C]^c [D]^d} = \frac{1}{K}$$

BONUS QUESTION

Assessed on: Summer 2023, Final Exam



At equilibrium, you find that the concentrations are 0.28 M for N_2 and 0.22 M for H_2 . If $K = 0.78$, then what is the equilibrium concentration (in M) of NH_3 ? Assume no changes in temperature.